

number system

```
In [1]: bin(45)
```

```
Out[1]: '0b101101'
```

```
In [5]: bin(50)
```

```
Out[5]: '0b110010'
```

```
In [1]: 25
```

```
Out[1]: 25
```

```
In [6]: bin(48)
```

```
Out[6]: '0b110000'
```

```
In [7]: bin(40)
```

```
Out[7]: '0b101000'
```

```
In [4]: bin(25) #we convert numn=ber inti binary digit
```

```
Out[4]: '0b11001'
```

```
In [3]: bin(30)
```

```
Out[3]: '0b11110'
```

```
In [8]: bin(50)
```

```
Out[8]: '0b110010'
```

```
In [6]: int(0b11110)
```

```
Out[6]: 30
```

```
In [7]: int(0b11001)
```

```
Out[7]: 25
```

```
In [1]: oct(25) #convert number to octal number system
```

```
Out[1]: '0o31'
```

```
In [12]: int(25) #octal to number
```

```
Out[12]: 25
```

```
In [13]: hex(16)
```

Out[13]: '0x10'

In [14]: `int(0x10)`

Out[14]: 16

In [15]: `int(0x9)`

Out[15]: 9

In [16]: `int(0x19)`

Out[16]: 25

complement operator

In [18]: `~12`

Out[18]: -13

In [19]: `~35`

Out[19]: -36

In [20]: `~19`

Out[20]: -20

In [22]: `12 & 13`

Out[22]: 12

In [23]: `12 | 13`

Out[23]: 13

In [24]: `12 ^ 13`

Out[24]: 1

In [25]: `12 << 2`

Out[25]: 48

In [26]: `12 >> 1`

Out[26]: 6

In [27]: `12 >> 2`

Out[27]: 3

```
In [28]: #bitwise completed
```

```
In [2]: int(0b11001)
```

```
Out[2]: 25
```

```
In [2]: bin(30)
```

```
Out[2]: '0b11110'
```